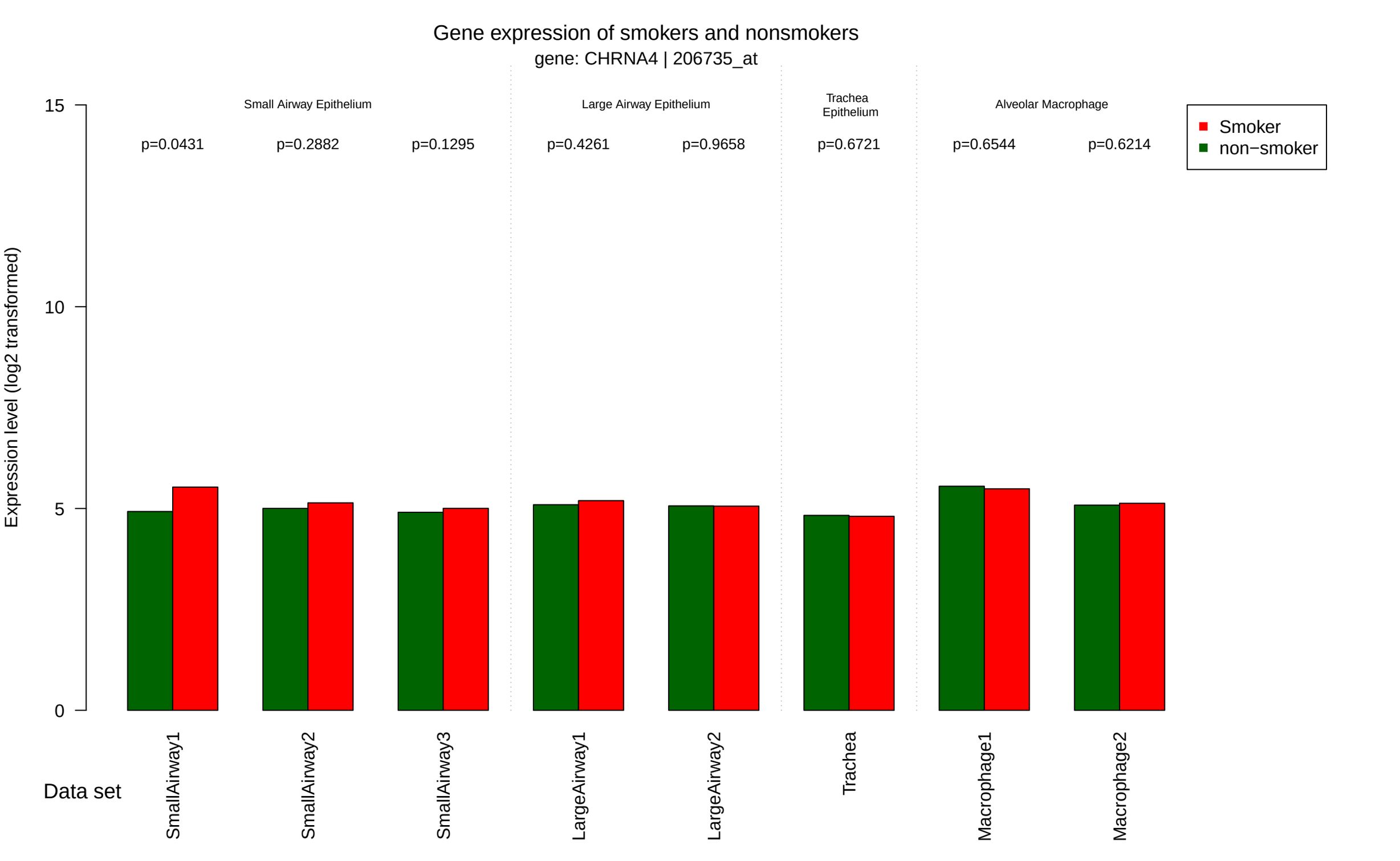
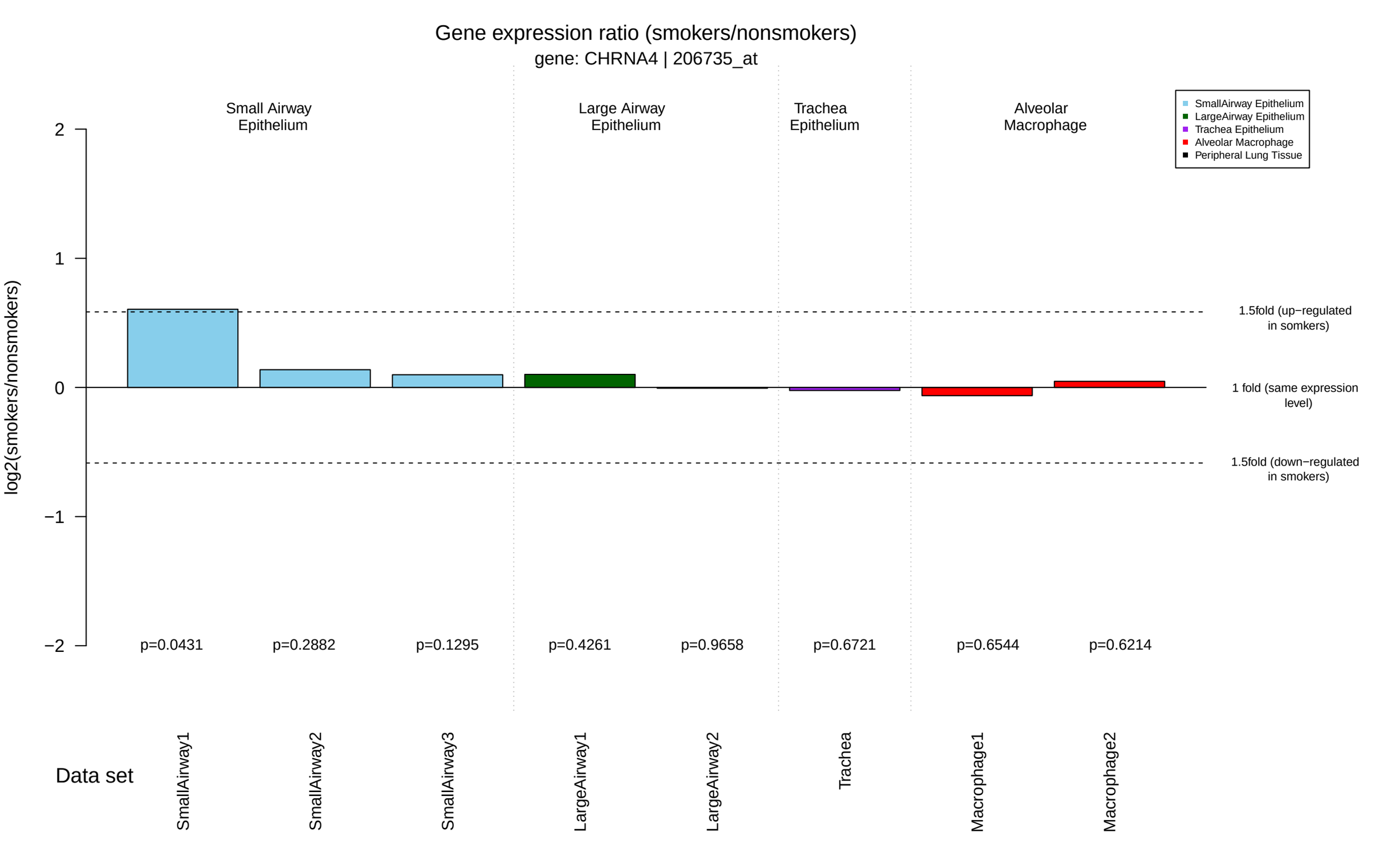
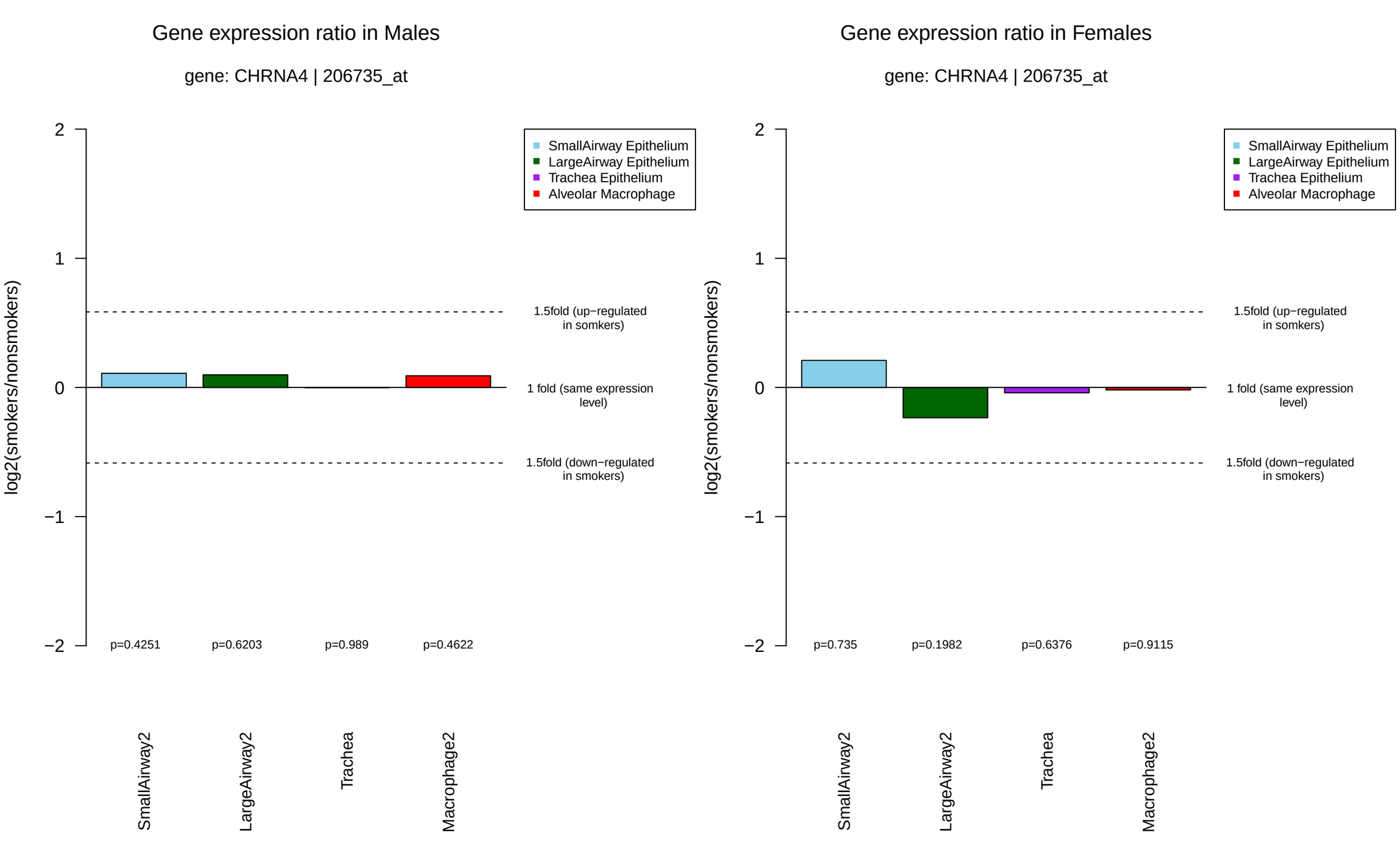
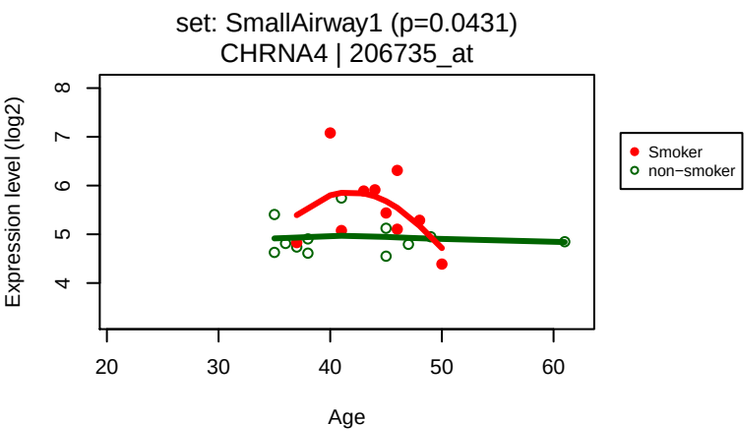
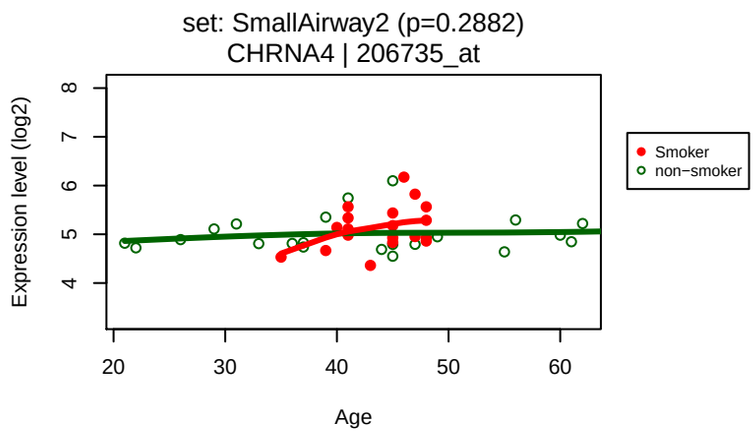
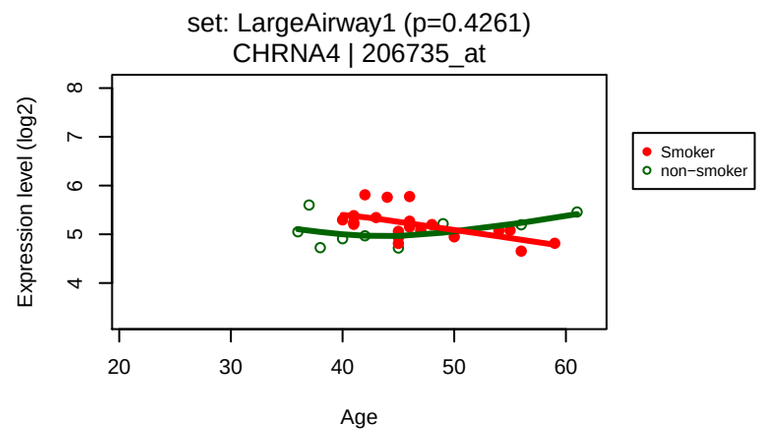

Gene expression of smokers and nonsmokers
gene: CHRNA4 | 206735_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0431
p=0.2882
p=0.1295
p=0.4261
p=0.9658
p=0.6721
p=0.6544
p=0.6214
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

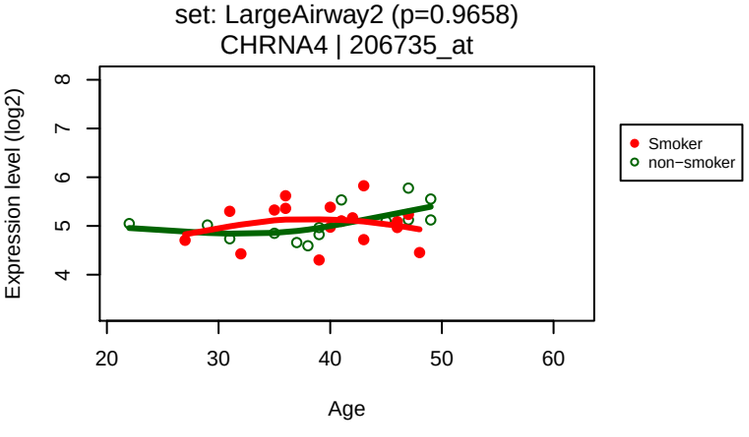

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA4 | 206735_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0431
p=0.2882
p=0.1295
p=0.4261
p=0.9658
p=0.6721
p=0.6544
p=0.6214
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

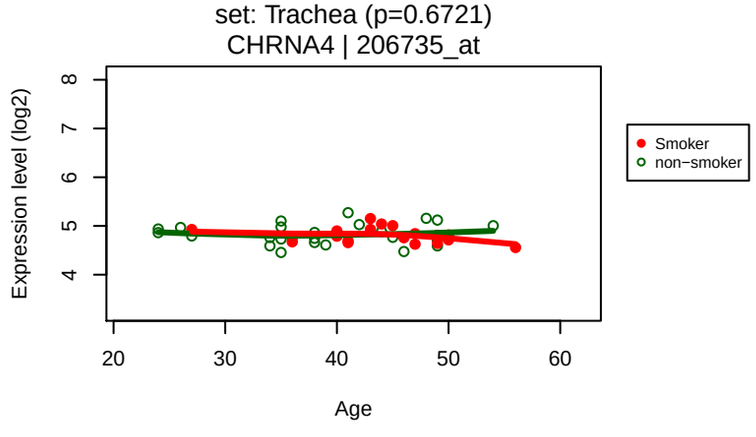

Gene expression ratio in Males
gene: CHRNA4 | 206735_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4251
p=0.6203
p=0.989
p=0.4622
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA4 | 206735_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.735
p=0.1982
p=0.6376
p=0.9115
SmallAirway2
LargeAirway2
Trachea
Macrophage2

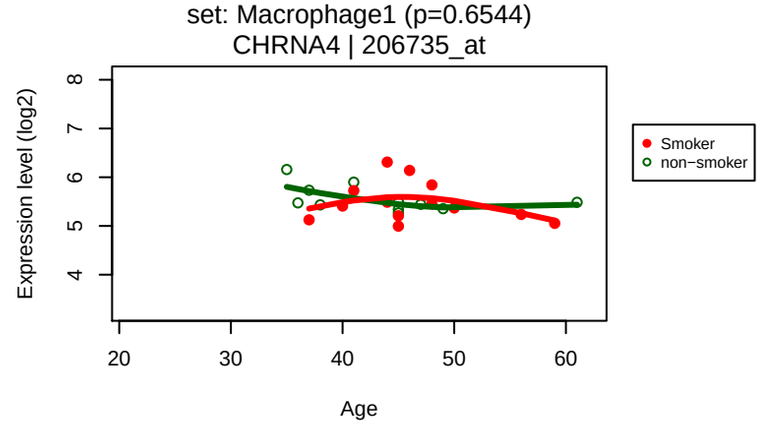

set: SmallAirway1 (p=0.0431)
CHRNA4 | 206735_at
Expression level (log2)
Age
Smoker
non−smoker

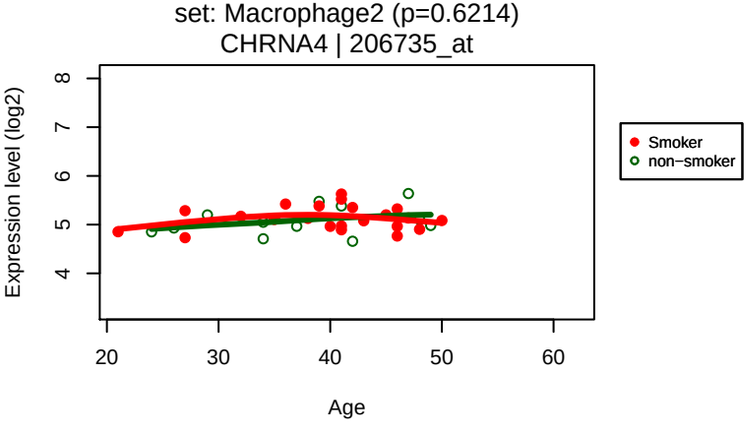

set: SmallAirway2 (p=0.2882)
CHRNA4 | 206735_at
Expression level (log2)
Age
Smoker
non−smoker

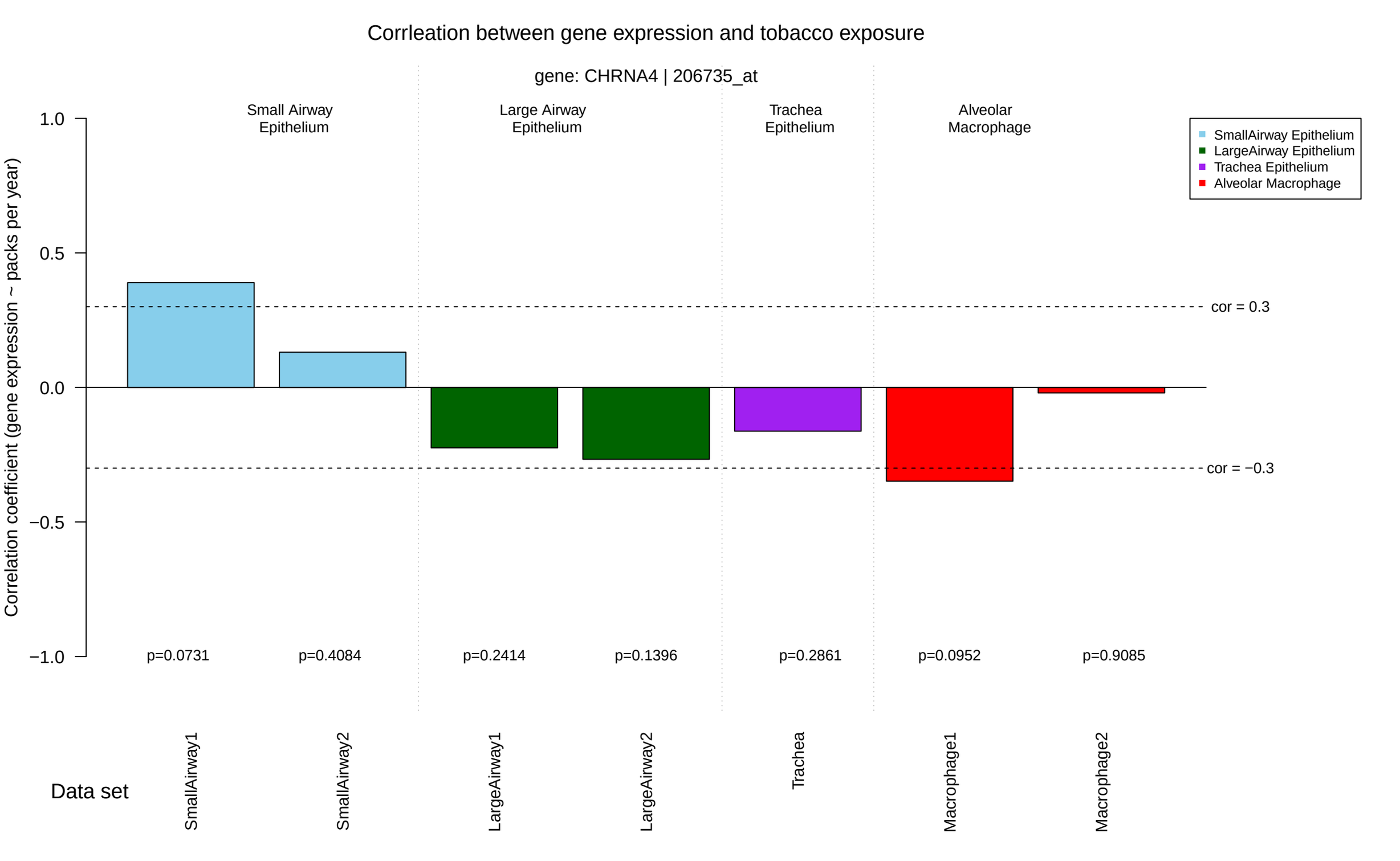

set: LargeAirway1 (p=0.4261)
CHRNA4 | 206735_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9658)
CHRNA4 | 206735_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6721)
CHRNA4 | 206735_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6544)
CHRNA4 | 206735_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6214)
CHRNA4 | 206735_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA4 | 206735_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0731
p=0.4084
p=0.2414
p=0.1396
p=0.2861
p=0.0952
p=0.9085
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2